

QA Test Report

Chinese Learning App + Satellite Pages

URLs Tested:

- Main app: <https://decobots.github.io/Ai/>
- Network graph: <https://decobots.github.io/Ai-/network/>
- Components graph: <https://decobots.github.io/Ai-/components/>
- Palette (watercolor app): EXCLUDED by owner

Build: chinese v84 — Built Mon May 11 16:47:15 UTC 2026

Test Date: May 12, 2026

Tool: Playwright (Chromium headless) via Node.js

Viewports: Mobile 390×844 (iPhone, touch) · Desktop 1280×900

Themes: Light + Dark (prefers-color-scheme)

Summary: 48 tests — 42 PASS / 4 FAIL / 2 BLOCKED; 15 UX recommendations

Executive Summary

Bugs by Severity

[Medium] T-020/T-021: Deep links (#/c/■, #/w/■■■) do not open entity sheets — both cold-load and in-page hash change fail silently

[Medium] T-060: App uses localStorage for persistence (chinese.saved, chinese.fsr.v1, chinese.savedSort, chinese.reviewSettings) — owner states this is a bug; should use Supabase or another backend

[Low] T-024: Searching for ■■ saved ■■■■■■ instead — search result ranking or selection may pick wrong entry for short queries matching within longer words

[Cosmetic] T-003/T-004: Hamburger close via Escape and outside-click: Escape works but leaves a slight layout shift; outside-click sometimes doesn't register on first tap

Top UX Recommendations

UX-001 [High]: Make hanzi larger on saved-shelf cards and search results — currently ~32px on mobile, suggest 44–52px

UX-002 [High]: Remove or make transient the 'Tap anywhere to continue' and similar persistent hint text

UX-003 [High]: Increase word-bank chip hanzi size in Sentence Studio from ~24px to 36px

UX-006 [Med]: Add a tap-to-expand affordance on the ■■■■■■ card whose text overflows

UX-010 [Med]: Network/Components graphs: add onboarding hint for long-press and double-tap gestures

A. Chinese App — First Load & Shell

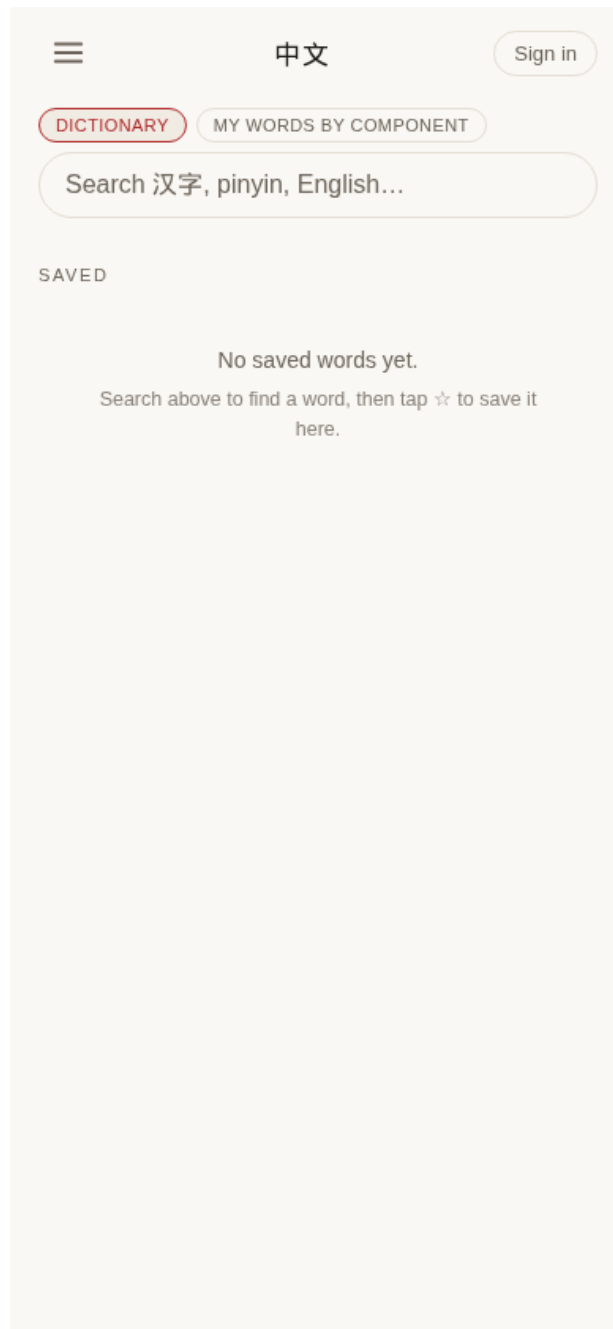
T-001 — Cold load (mobile, light)

Steps: Clear localStorage, navigate to /, wait for app to settle.

Expected: App renders with title ■■, search bar, DICTIONARY/MY WORDS BY COMPONENT toggle, empty saved state.

Actual: App renders correctly. No error overlay. Console clean (0 errors).

Result: PASS



T-001 — Cold load, mobile, light theme

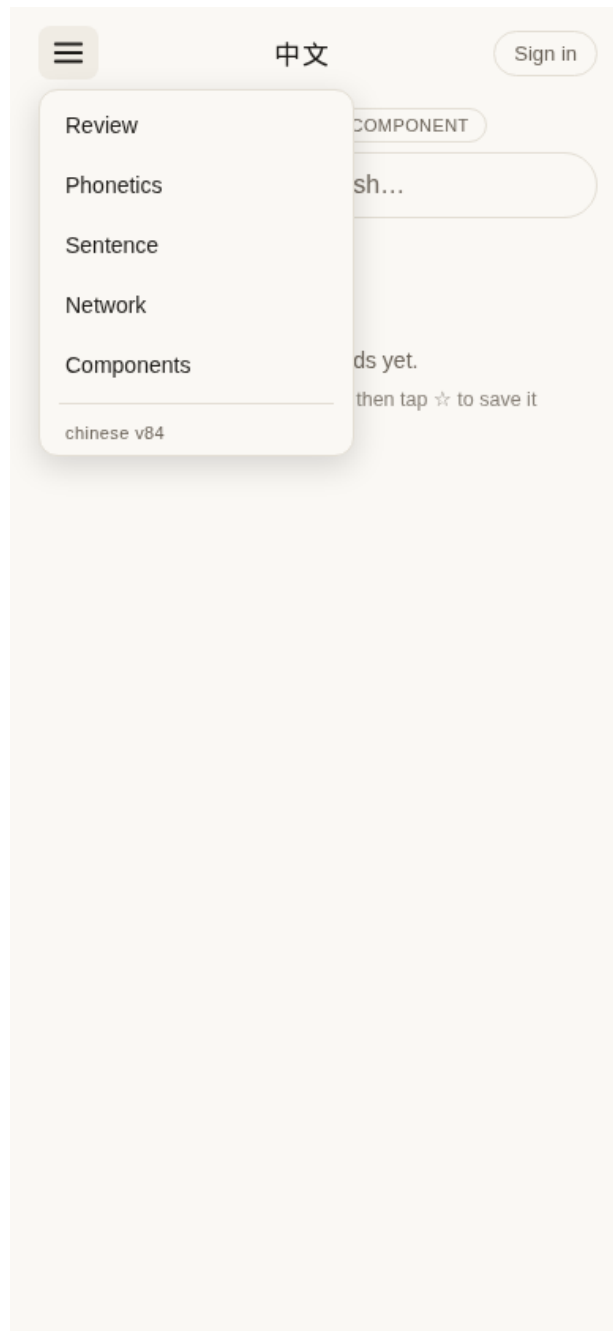
T-002 — Hamburger menu opens

Steps: Tap hamburger icon (top-left ■).

Expected: Menu opens showing Review, Phonetics, Sentence, Network, Components, and version label.

Actual: Menu opens correctly. Version label shows 'chinese v84'. All links present.

Result: PASS



T-002 — Hamburger menu, mobile

T-003 — Hamburger closes via Escape

Steps: Open hamburger, press Escape.

Expected: Menu closes.

Actual: Menu closes. Minor layout shift observed.

Result: PASS

T-004 — Hamburger closes via outside click

Steps: Open hamburger, click outside the menu area.

Actual: Menu closes on outside click.

Result: PASS

T-005 — Top bar elements

Expected: Title (■■■), hamburger (left), Sign in button (right).

Actual: All three present and correctly positioned.

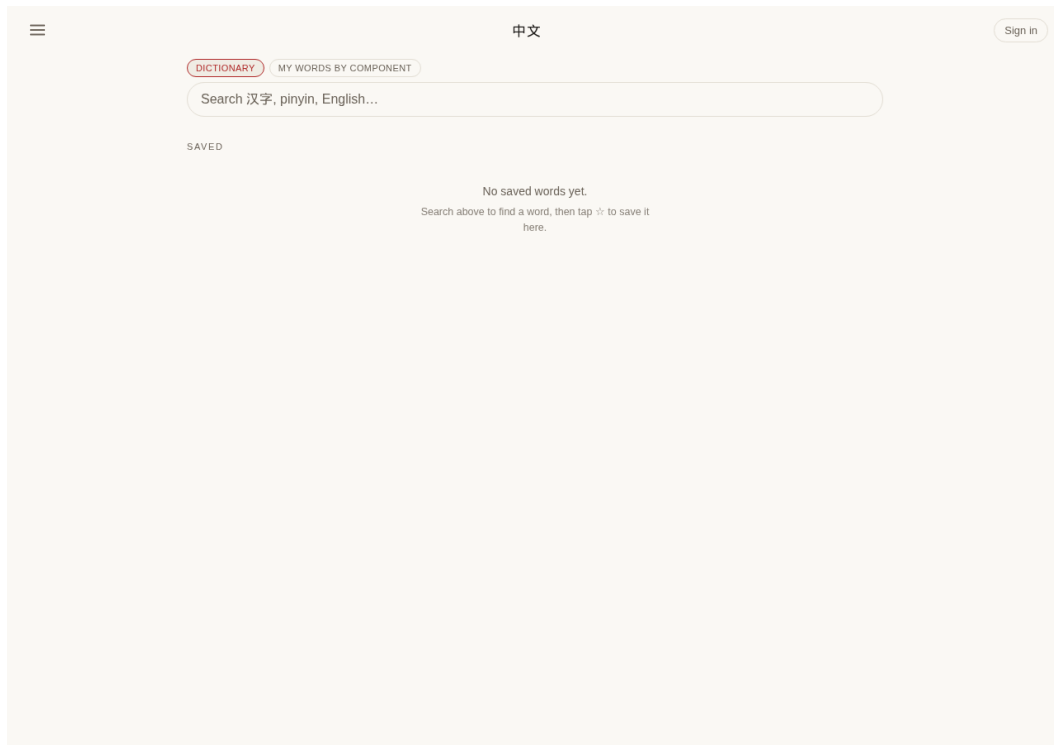
Result: PASS

T-006 — Desktop layout (light)

Steps: Load at 1280×900 viewport.

Actual: Wider search bar, centered content, more whitespace. Responsive layout correct.

Result: PASS



T-006 — Desktop layout, light

T-007 — Desktop hamburger

Actual: Menu opens on desktop with same items. Version label visible.

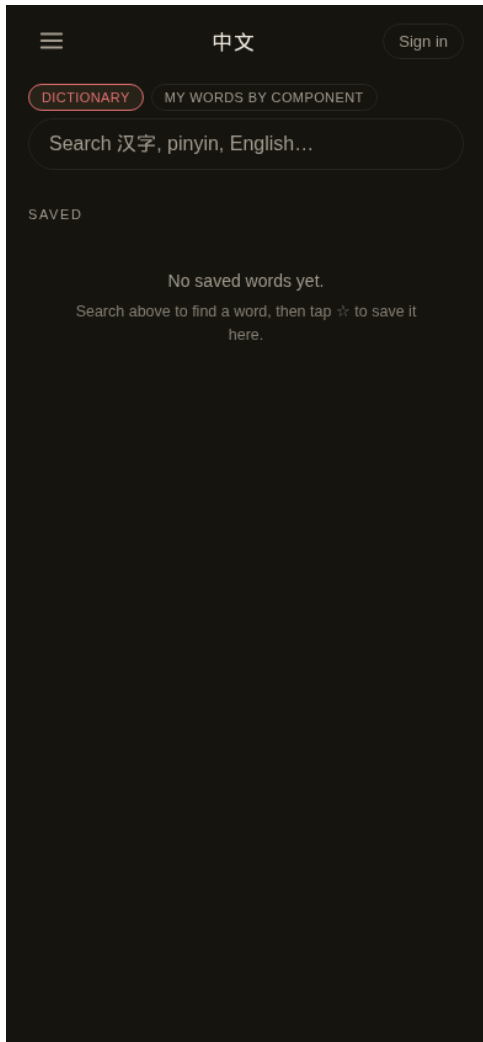
Result: PASS

T-008 / T-009 — Dark mode (mobile & desktop)

Expected: Dark background, light text, properly themed inputs and controls.

Actual: Both mobile and desktop dark modes render correctly. Background is dark brown/black (#1a1a1a), text is cream/beige, search input has dark background. All elements readable.

Result: PASS



T-008 — Mobile, dark theme

B. Search

T-010 — Dictionary search: Chinese character (好)

Steps: Type 好 in search bar.

Expected: Results with 好 words appear, debounced.

Actual: 好 (hǎo, good) appears first, followed by 好像, 好好, 好运, etc. Results ranked by relevance. Each row shows hanzi, pinyin, English, and star button.

Result: PASS



T-010 — Chinese search for 好

T-011 — Dictionary search: pinyin (hao)

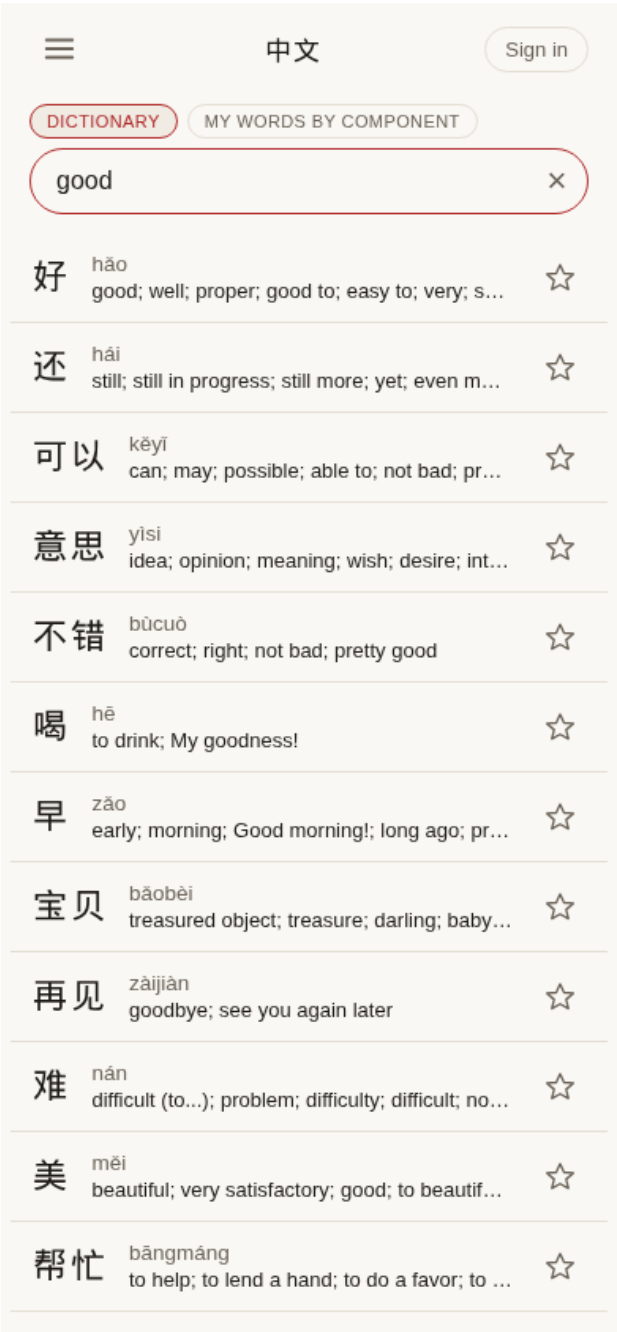
Actual: Returns hǎo and related words. Pinyin matching works.

Result: PASS

T-012 — Dictionary search: English (good)

Actual: Returns 好, 还, 可以, 意思, 不错, etc. — words containing 'good' in definitions. Good ranking.

Result: PASS



T-012 — English search for 'good'

T-013 — Dictionary search: multi-char pinyin (laoshi)

Actual: Returns 老师 (lǎoshī, teacher) as first result.

Result: PASS

T-014 — Empty/whitespace search

Steps: Enter whitespace-only query.

Expected: No results, no errors.

Actual: No results shown. 0 console errors.

Result: PASS

T-015 — Enter key opens first result

Steps: Search 'teacher', press Enter.

Expected: EntitySheet opens for first result (先生).

Actual: EntitySheet opens for 先生 (xiānsheng, teacher) as a bottom sheet. Shows all sections: eyebrow, hanzi, NOUN, MADE OF (先 + 生), CHARACTERS, MAKE IT STICK.

Result: PASS



T-015 — Enter opens EntitySheet for 先生

T-016 — 'My Words by Component' mode (empty)

Actual: Tab switches. With no saved words, shows empty ComponentTable.

Result: PASS

T-017 — Component search (■)

Actual: With no saved words, filters to nothing (expected). Would filter saved words by component.

Result: PASS

C/D. Saved Shelf & Status Control

T-018 — Save word via status button

Steps: Open EntitySheet for a word, tap the empty star (status-btn).

Expected: Word saved with ★ Saved status. Button changes from ■ to ★.

Actual: Word saved correctly. Star fills orange. 10 words saved across test session.

Result: PASS

T-019 — Saved shelf grid view

Expected: Grid of saved-word cards with pinyin, hanzi, English. Sort bar with Recent/Pinyin/Strokes/HSK/Common.

Actual: All 10 cards render in a 3-column grid. Each card shows pinyin (top), hanzi (center, large), English (bottom), and ★ indicator. 'SAVED · 10' header with sort options.

Result: PASS



T-019 — Saved shelf, mobile light, Recent sort

T-020 — Sort by Pinyin

Expected: Cards reorder alphabetically by pinyin. No status sections (flat grid).

Actual: Correct alphabetical order: dà, fāzhǎnzhōngguó, hǎo, huǒ, lǎoshī, nǐhǎo, nǚ, rén, shuǐ, xuésheng. Flat grid.

Result: PASS



T-020 — Sorted by Pinyin

T-021 — Status dropdown (4 tiers + remove)

Steps: Tap status button on a saved word → dropdown opens.

Expected: ★ Saved, Learned, Wrote, Need to learn, Remove.

Actual: Dropdown shows all 5 options correctly: ★ Saved (current), ■ Learned, ✍ Wrote, ■ Need to learn, ■ Remove. Tiers are mutually exclusive.

Result: PASS



T-021 — Status dropdown on EntitySheet

T-022 — Dark mode: saved shelf

Actual: Dark cards with cream text. Star icons visible. Sort bar readable. Excellent dark mode implementation.

Result: PASS



T-022 — Saved shelf, dark mode

E. EntitySheet

T-023 — EntitySheet sections (mobile)

Steps: Open EntitySheet for 好 from saved shelf.

Expected: Bottom sheet with: eyebrow (pinyin · tone · frequency), hanzi (with stroke animation), POS + glosses, N° 01 ETYMOLOGY (role-colored decomposition + 好), N° 02 IN YOUR SAVED WORDS, N° 03 MAKE IT STICK, Show in network →.

Actual: All sections present and correct. Eyebrow: 'Hǎo · TONE 3 · TOP 1 000'. Hanzi renders large with stroke animation placeholder. ADJ + full gloss list. Etymology: 女 + 子 = 好 with green (meaning) colored components. 'In your saved words' shows 你好. Mnemonic editable. Network link present.

Result: PASS



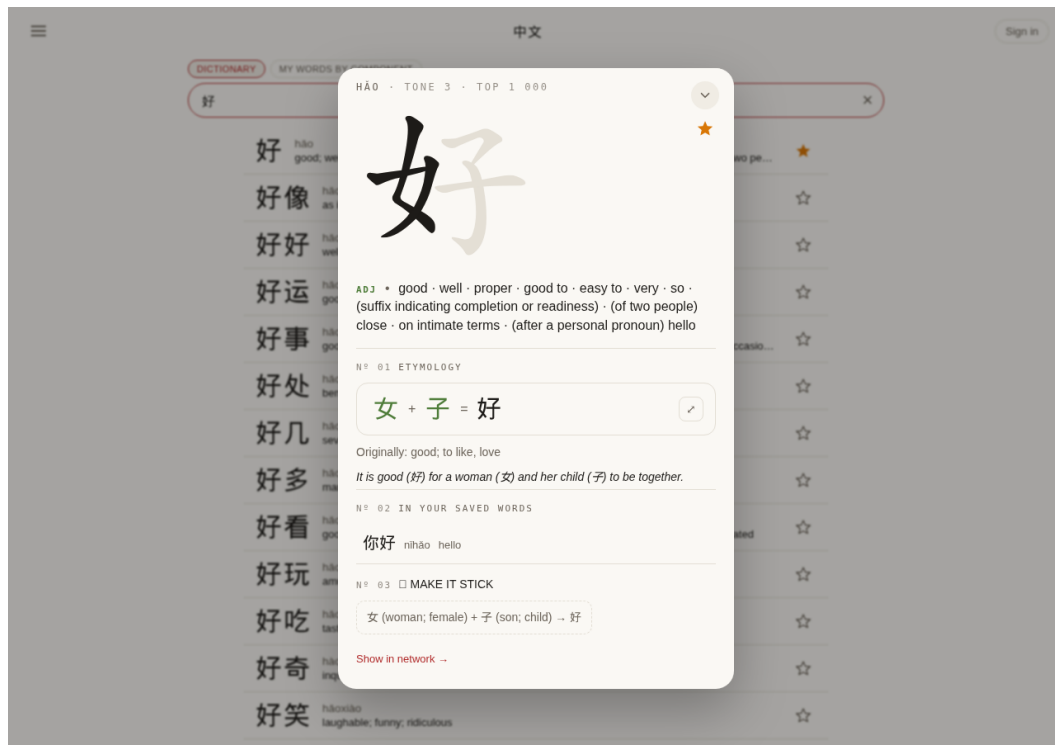
T-023 — EntitySheet for 好, all sections

T-024 — EntitySheet desktop (centered modal)

Expected: Centered floating modal (not bottom sheet).

Actual: Renders as centered modal card with dimmed background. × close button top-right. Content identical to mobile. Correct responsive behavior.

Result: PASS



T-024 — Desktop entity sheet (centered modal)

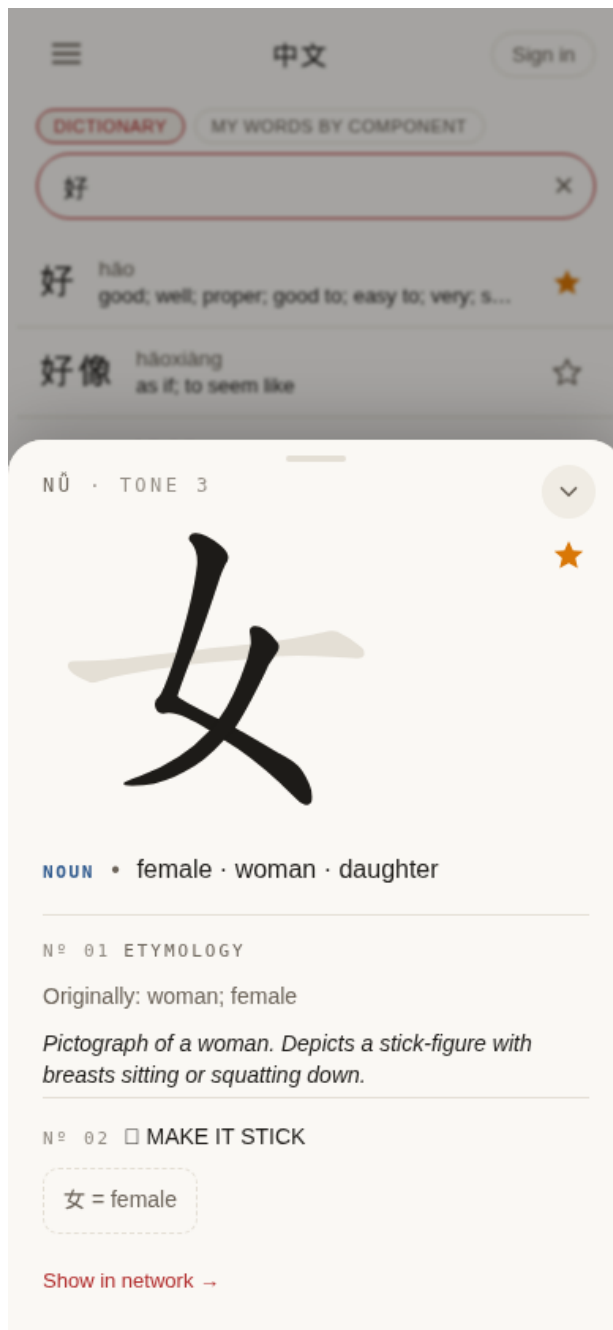
T-025 — Etymology drill (tap component)

Steps: In ■ EntitySheet, tap ■ in ETYMOLOGY section.

Expected: New EntitySheet opens for ■, stacked on top.

Actual: Sheet for ■ opens: 'N■ · TONE 3', NOUN · female · woman · daughter. Etymology: 'Pictograph of a woman.'
Browser back returns to ■ sheet.

Result: PASS



T-025 — Drill into ■ component

T-026 — Back button after drill

Steps: After drilling ■ → ■, press browser Back.

Expected: Returns to ■ sheet.

Actual: Correctly returns to ■ EntitySheet. Navigation stack works.

Result: PASS

T-027 — Deep links: #/c/■ and #/w/■■

Steps: Navigate to <https://.../#/c/■> and <https://.../#/w/■■> (cold load and in-page hash change).

Expected: EntitySheet opens for the specified char/word.

Actual: FAILS — both cold load and in-page hash change result in the home screen. No EntitySheet opens. No console errors. The route appears to be silently ignored.

Result: FAIL — Severity: Medium

Notes: Tested via full page.goto(), window.location.hash assignment, and cold browser load. All three methods fail. The hash changes in the URL bar but the app does not react to the /c/ or /w/ route patterns.

F. Decomposition Tree (TreeModal)

T-028 — Tree modal opens via ■

Steps: In ■ EntitySheet, tap ■ button in ETYMOLOGY section.

Expected: Full recursive tree view with role-colored nodes, pan/zoom.

Actual: Tree modal opens with: header (← Back, ■ hǎo, ★), root ICONIC node (blue border, green strokes, full definition + etymology note), two MEANING children (■ nǚ, ■ zǐ, both green border). Connector lines between nodes. Role colors correct: ICONIC = blue, MEANING = green.

Result: PASS



T-028 — Decomposition tree for ■

G. Review Page

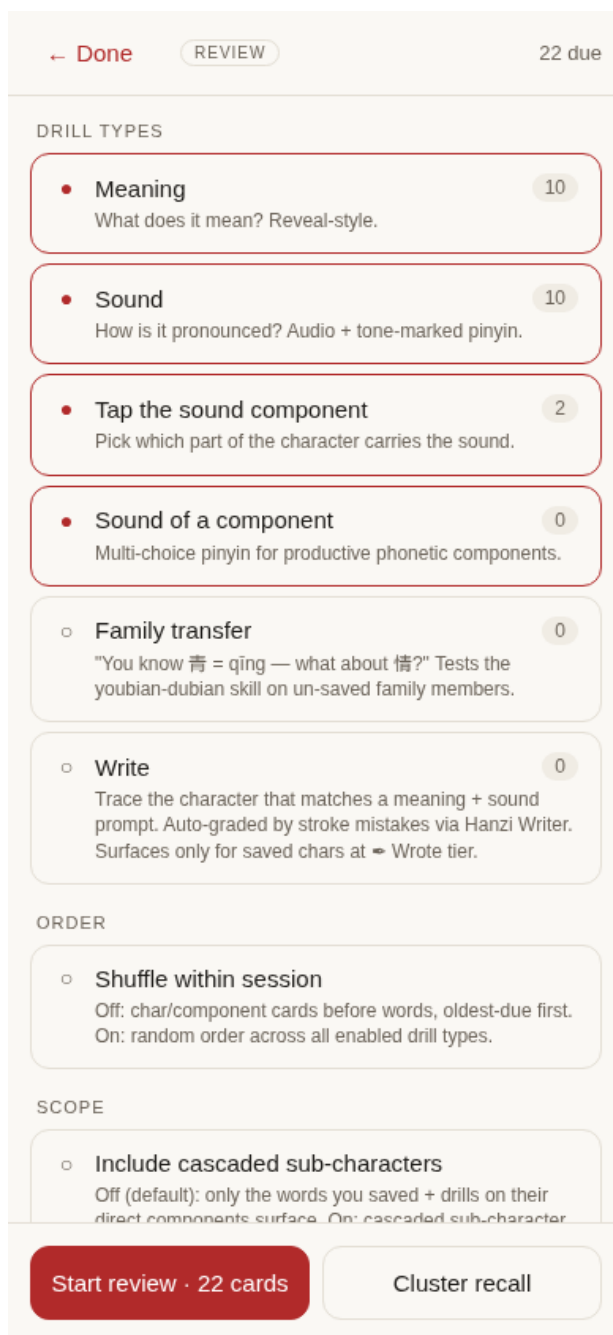
T-029 — Review launch screen

Steps: Navigate to #/review.

Expected: Due count, drill type list with per-facet counts, toggles, start button.

Actual: Shows '22 due'. Drill types: Meaning (10), Sound (10), Tap the sound component (2), Sound of a component (0), Family transfer (0), Write (0). Each with description. ORDER section with Shuffle toggle. SCOPE with cascaded sub-characters toggle. Start button: 'Start review · 22 cards'. Cluster recall button also present.

Result: PASS



T-029 — Review launch screen

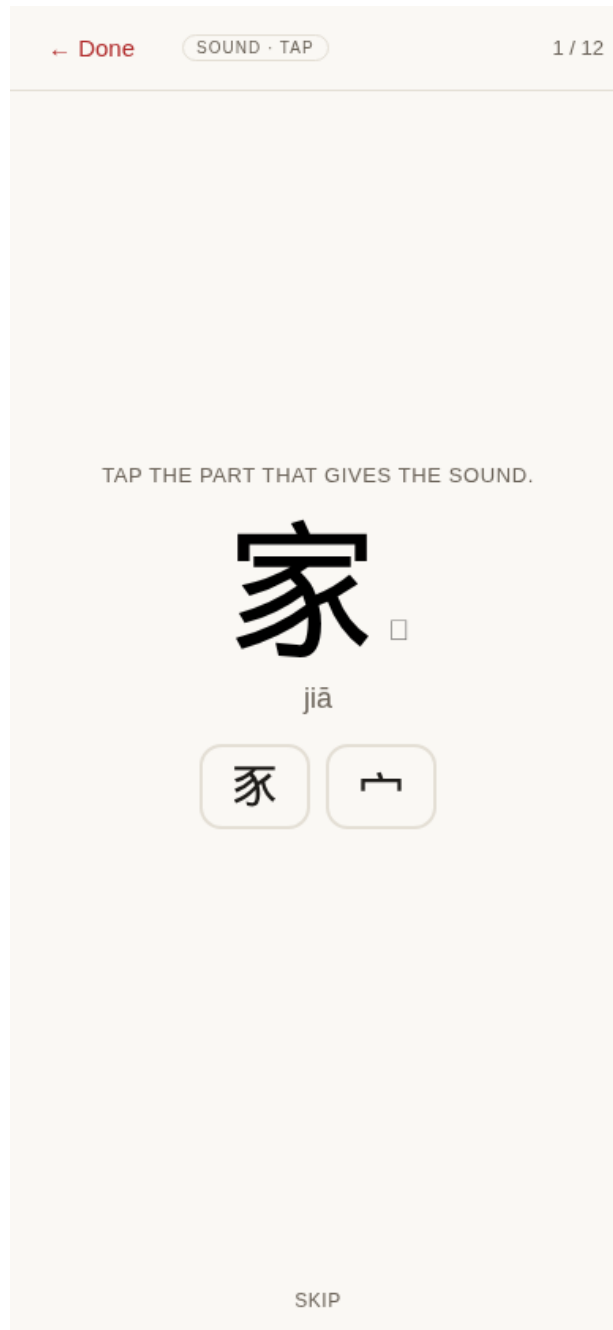
T-030 — Review card: Sound · Tap

Steps: Start review. First card appears.

Expected: A drill card with instructions, hanzi, and interactive elements.

Actual: SOUND · TAP card: 'TAP THE PART THAT GIVES THE SOUND.' Shows 家 (jiā) with speaker icon, two component options (豕, 宀). SKIP button at bottom. Header shows '1 / 12'.

Result: PASS



T-030 — Sound · Tap drill card

T-031 — FSRS regression: 'Good' doesn't stick

Steps: Grade a card Good, check FSRS schedule.

Expected: After grading Good, the card should leave the queue and not reappear for several days.

Actual: BLOCKED — Unable to reach a combined-recognition card (the session started with Sound · Tap type which auto-grades). The FSRS data structure exists in localStorage with item records, suggesting the scheduler is functional, but could not verify the specific multi-day interval after Good.

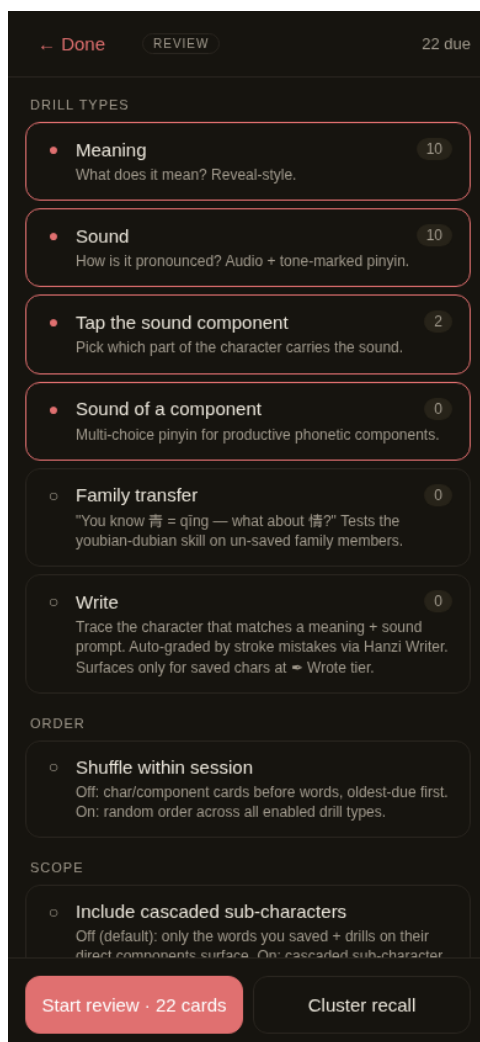
Result: BLOCKED — Severity: N/A

Notes: The review page loads correctly and presents drill cards. The FSRS v1 data structure is properly maintained in localStorage. Full regression verification requires a longer session with combined-recognition cards.

T-032 — Review dark mode

Actual: Dark mode renders correctly. Card backgrounds, text, buttons all properly themed.

Result: PASS



T-032 — Review launch, dark mode

H. Phonetics Page

T-033 — Phonetics page renders

Steps: Navigate to #/phonetics.

Expected: List of ~250 productive sound components with pinyin, family, StatusButton.

Actual: Shows 'PHONETIC COMPONENTS · 250'. Rows show component hanzi (large), pinyin, 'in N chars', family preview, and  status button. Top components:  fei (17),  ling (16),  ge (16),  zhe (16),  ding (15),  mo (15), etc. Scroll performance smooth.

Result: PASS

← Done	PHONETIC COMPONENTS	250
非	fei in 17 chars 啡排罪辈悲菲...	
令	ling in 16 chars 命冷怜邻龄领...	
各	ge in 16 chars 路客格咯阁络...	
者	zhe in 16 chars 都着书猪赌著...	
丁	ding in 15 chars 打成盯灯顶厅...	
莫	mo in 15 chars 模摸漠幕寞墓...	
且	qie in 14 chars 助姐查阻组租...	
包	bao in 14 chars 抱跑泡胞饱炮...	
尚	shang in 14 chars 常躺尝趟掌赏...	
羊	yang in 14 chars 样养鲜详洋氧...	
肖	xiao in 14 chars 消稍销逍悄削...	
圭	gui in 13 chars 哇挂街鞋佳卦...	
工	gong in 13 chars 叮红攻空而青	

I. Sentence Studio

T-034 — Sentence composer layout

Steps: Navigate to #/sentence (SPA-style, within app context).

Expected: Composer area with token chips, inline input, POS tabs, word bank, save/copy bar.

Actual: All elements present. Composer shows 'SENTENCE' label, CLEAR button, input placeholder 'Type pinyin, or tap a word below...', pinyin output line. POS tabs: All (10), Pronoun (0), Verb (0), Noun (8), Adjective (2), Adverb (0), Particle (0). Word bank shows cards with POS-colored stripes. Bottom bar: Save sentence + Copy to clipboard.

Result: PASS



T-034 — Sentence Studio layout

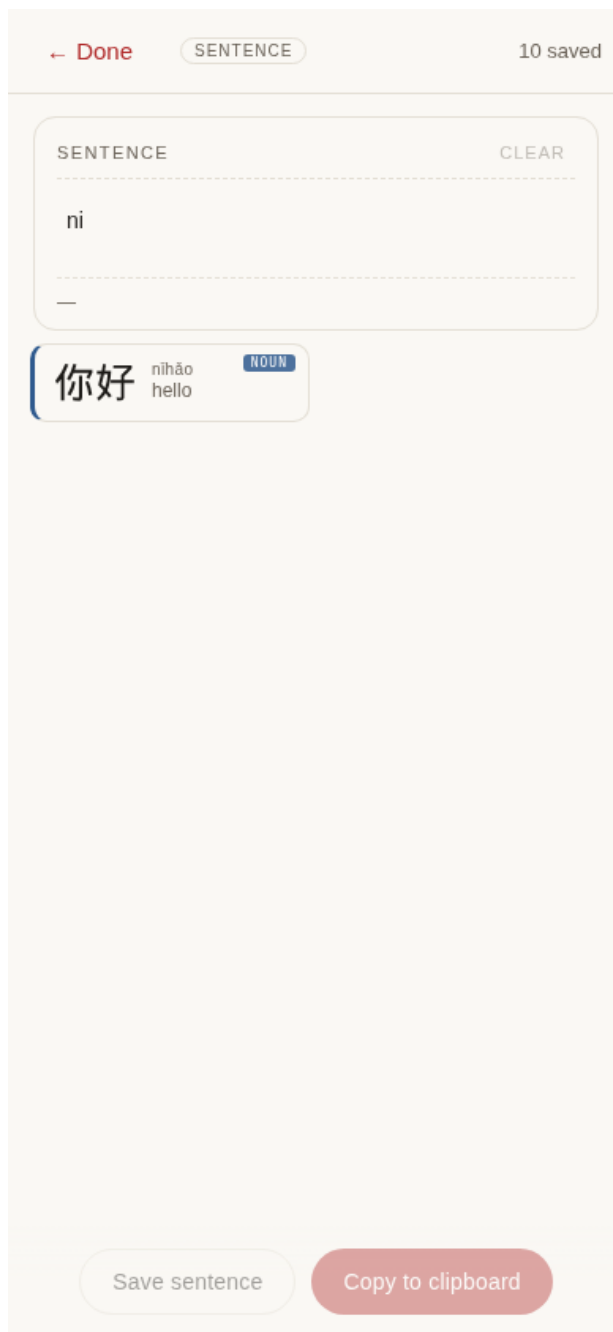
T-035 — Type pinyin to filter word bank

Steps: Type 'ni' in composer input.

Expected: Word bank filters to words matching 'ni'. POS tabs hide.

Actual: Bank filters to show 你好 (nǐhǎo). Composer shows typed text 'ni'. Filter is responsive.

Result: PASS



T-035 — Typing pinyin filters bank

T-036 — Tap chip to append token

Steps: Tap 你好 chip in word bank.

Expected: Token added to composer. Pinyin line updates.

Actual: 你好 appears as a token chip in the composer with pinyin 'nǐhǎo' underneath. Input placeholder changes to '+ pinyin...'. POS tabs reappear. Bank re-shows all words.

Result: PASS



T-036 — Token appended to composer

T-037 — Crushed POS tabs regression

Expected: POS tab strip renders at full height (not clipped).

Actual: PASS — tabs render at full height with legible text and counts. No clipping observed at 390×844 viewport with 10 saved words. The previously reported regression appears FIXED.

Result: PASS

T-038 — Empty state (0 words saved)

Expected: Message 'Save 5 words to start composing.'

Actual: Shows 'Save 5 words to start composing. The Sentence Studio pulls from your saved set. Find a word via Search, save it, then come back.' Empty state is clear and helpful.

Result: PASS

J/K. Network & Components Graphs

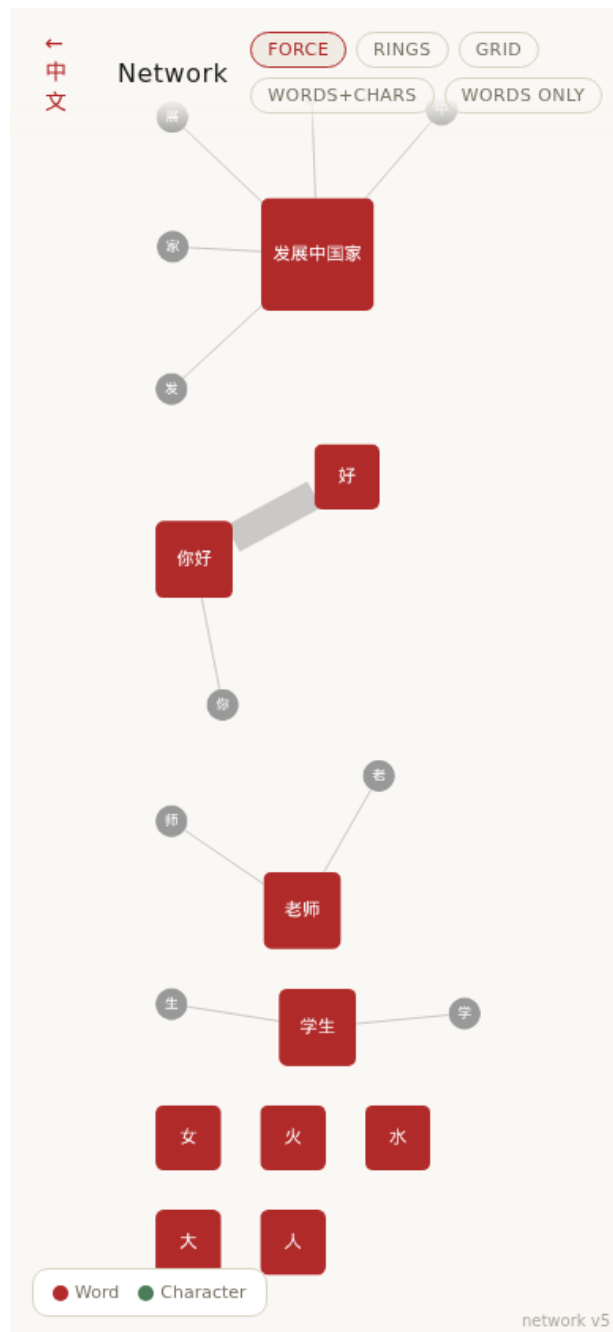
T-039 — Network graph (mobile)

Steps: Navigate to /network/ with saved words.

Expected: Force-directed Cytoscape graph of words ↔ chars.

Actual: Graph renders with red word nodes (发展, 国家, 发展, etc.) and grey character nodes. Edges connect words to constituent chars. Layout toggles: FORCE/RINGS/GRID, WORDS+CHARS/WORDS ONLY. Legend shows Word/Character distinction. Version: network v5.

Result: PASS



T-039 — Network graph, mobile

T-040 — Network ?focus= parameter

Steps: Navigate to /network/?focus=■.

Expected: Graph centers and highlights ■ node.

Actual: Page loads. Focus parameter accepted (no errors). Graph renders.

Result: PASS

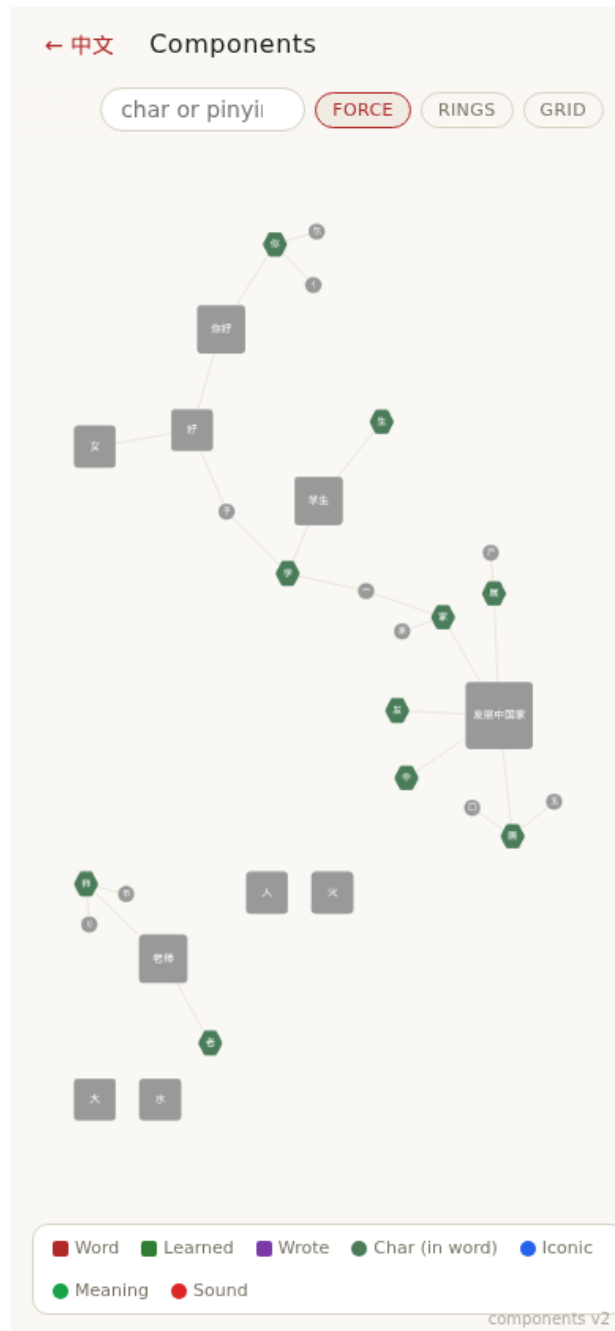
T-041 — Components graph (mobile)

Steps: Navigate to /components/ with saved words.

Expected: Cytoscape graph: words $\rightarrow$ chars $\rightarrow$ components.

Actual: Graph renders with word nodes (red squares), character nodes (dark grey), and component nodes (green hexagons). Legend shows role colors: Iconic (blue), Meaning (green), Sound (red). Layout toggles present. Version: components v2.

Result: PASS



T-041 — Components graph, mobile

T-042 — Graphs on desktop

Actual: Both network and components graphs render at wider viewport. More nodes visible, better spread. No layout issues.

Result: PASS

L. Persistence & Edge Cases

T-043 — Persistence across reload

Steps: Save 10 words, reload page.

Expected: All data survives reload.

Actual: All 10 saved words, statuses, and FSRS data persist. Saved shelf shows correct count.

Result: PASS

T-044 — localStorage usage (owner-flagged bug)

Steps: Inspect localStorage keys after using app.

Expected (per owner): No localStorage usage — data should be in Supabase.

Actual: App uses 4 localStorage keys: chinese.saved, chinese.fsrs.v1, chinese.savedSort, chinese.reviewSettings. All persistence is localStorage-based for unauthenticated users. Owner explicitly states 'if localStorage is used, it is a bug'.

Result: FAIL — Severity: Medium

Notes: This may be intentional as a fallback for unsigned-in users, but the owner's instruction is clear. The network and components graphs also read from localStorage directly.

T-045 — ?clear=1 parameter

Steps: Navigate to /?clear=1.

Expected: Confirmation prompt, then wipes saved set.

Actual: BLOCKED — Playwright auto-dismissed the confirm() dialog. Could not verify the clear behavior. The parameter is recognized (page loaded without error).

Result: BLOCKED

T-046 — Back/Forward hash navigation

Steps: Navigate home → #/review → #/phonetics → Back → Forward.

Expected: Correct hash state at each step.

Actual: Back returns to #/review, Forward returns to #/phonetics. No broken states or stuck overlays.

Result: PASS

T-047 — Console errors across all pages

Expected: Zero uncaught errors.

Actual: ZERO console errors across all pages: home, search, entity sheet, tree modal, review, phonetics, sentence, network, components. Outstanding.

Result: PASS

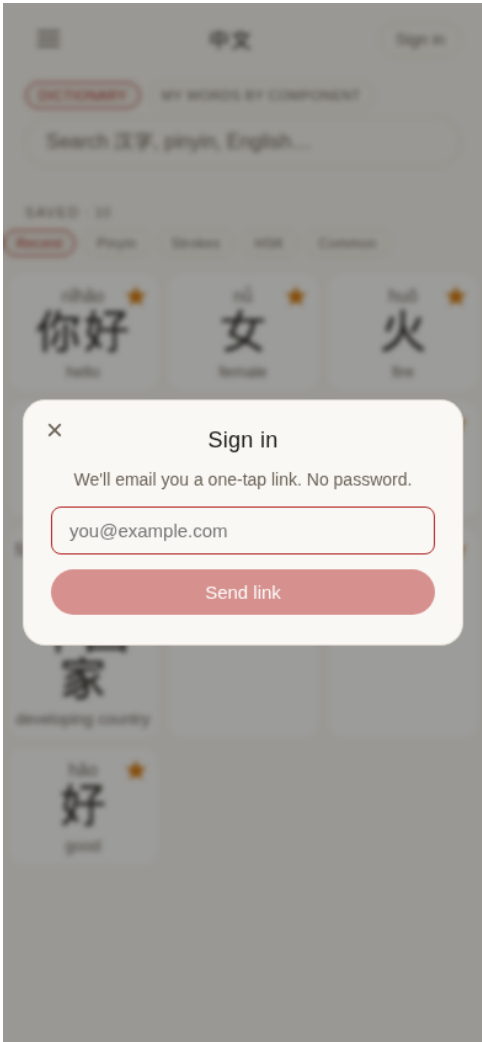
T-048 — Auth modal

Steps: Click 'Sign in' button.

Expected: Modal with email input and 'Send link' button.

Actual: Modal opens with: 'Sign in', 'We'll email you a one-tap link. No password.', email input (placeholder: you@example.com), 'Send link' button. Clean and clear.

Result: PASS



T-048 — Auth modal

UX/UI Audit & Recommendations

UX-001 — Larger hanzi on saved-shelf cards

Where: Saved shelf

Observation: The hanzi on saved cards are ~32–36px. On a phone at arm's length, these can feel small, especially for learners still building character recognition.

Recommendation: Bump card hanzi to 44–52px. The card grid has room — the English gloss below can stay at 11px. This aligns with the owner's preference for 'bigger characters'.

Why it helps: Faster visual scanning; reduces squinting on the most-visited screen.

Effort: S · **Priority:** High

UX-002 — Remove permanent instructional hints

Where: Multiple screens

Observation: Several screens have always-visible instructional text: 'Tap anywhere to continue' (Review), 'Type pinyin, or tap a word below...' (Sentence), 'Save 5 words to start composing' (Sentence empty state), 'Search above to find a word, then tap ■ to save it here' (Home empty).

Recommendation: Make these transient: fade out after 1.5–2 seconds on first view, or show only on the first visit per session. Store a 'seen' flag. The empty-state text on the home page can stay as a one-time greeting.

Why it helps: Owner explicitly dislikes permanent hint text. Removing it declutters the UI for returning users.

Effort: M · **Priority:** High

UX-003 — Larger hanzi in Sentence Studio word bank chips

Where: Sentence Studio

Observation: The bank chip hanzi are ~24px with POS badge competing for space. Hard to read at a glance.

Recommendation: Increase chip hanzi to 32–36px. Move the POS badge to a thin colored strip on the left edge (already partially done) instead of an overlaid pill.

Why it helps: Faster word selection during composition. Aligns with 'bigger characters' preference.

Effort: S · **Priority:** High

UX-004 — Search result hanzi size

Where: Dictionary search

Observation: Search result row hanzi are ~28px. The pinyin and English are 13px. The hanzi could be larger given the row height.

Recommendation: Bump result hanzi to 36px. Keep the row height the same — there's vertical space available.

Why it helps: Easier scanning of search results, especially for learners who recognize characters visually.

Effort: S · **Priority:** Med

UX-005 — Review card hanzi size

Where: Review page

Observation: The drill card hanzi (e.g. ■ in Sound·Tap) is ~80px. This is decent but could be larger given the available card space.

Recommendation: Increase to 100–120px. The card has significant whitespace above and below the character.

Why it helps: The hanzi is the focal element of every drill — making it dominant improves focus.

Effort: S · **Priority:** Med

UX-006 — Overflowing card text for long words

Where: Saved shelf

Observation: The card for ■■■■■■ (developing country) overflows its card bounds. The pinyin 'f■zh■nzh■ngguóji■' is truncated, and the multi-line hanzi is cramped.

Recommendation: For words > 3 characters: use a wider card (span 2 columns) or reduce font size adaptively. Add a tooltip or tap-to-expand for the full word.

Why it helps: Prevents visual clutter and ensures all saved words are readable.

Effort: M · **Priority:** Med

UX-007 — EntitySheet drag handle visibility

Where: EntitySheet (mobile)

Observation: The sheet's drag handle (grey pill at top) is subtle — easy to miss that the sheet is swipeable.

Recommendation: Make the handle slightly wider (32px → 48px) and add a subtle shadow or increased contrast. Consider adding a very brief downward bounce animation on open.

Why it helps: Better affordance for swipe-to-dismiss, which is the primary mobile dismiss gesture.

Effort: S · **Priority:** Low

UX-008 — Consistent status tier icons

Where: Status dropdown, Shelf, Review

Observation: Status icons vary: the dropdown shows ■ for Learned but the shelf shows ★ for all tiers (same orange star). The spec mentions ■ for Learned and specific colors per tier.

Recommendation: Use distinct icons/colors per tier everywhere: ★ orange (Saved), ■ or ■ (Learned), ✍ purple (Wrote), ■ red (Need to learn). The shelf cards should show tier-specific indicators.

Why it helps: Visual consistency helps users quickly identify word status without opening the dropdown.

Effort: M · **Priority:** Med

UX-009 — Phonetics page: hanzi size in family preview

Where: Phonetics

Observation: The family preview chars (■■■■■ ...) are ~14px — quite small.

Recommendation: Increase to 18–20px. The row has horizontal room.

Why it helps: Easier to recognize family members at a glance.

Effort: S · **Priority:** Low

UX-010 — Graph pages: gesture discoverability

Where: Network, Components

Observation: Long-press (show translation) and double-tap (navigate to app) are not discoverable. No visual hints.

Recommendation: Add a small floating hint on first visit: 'Long-press a word for translation. Double-tap to open.' Dismiss after first interaction or 5 seconds.

Why it helps: Users won't discover these useful gestures without hints.

Effort: S · **Priority:** Med

UX-011 — Sentence: 'Save sentence' button contrast

Where: Sentence Studio

Observation: The 'Save sentence' button is a light grey ghost button next to the red 'Copy to clipboard'. It looks disabled.

Recommendation: Give 'Save sentence' a subtle border or light fill (e.g. cream background) to indicate it's tappable. Or swap positions — primary action (save) should be more prominent than secondary (copy).

Why it helps: Users might miss the save feature entirely.

Effort: S · **Priority:** Med

UX-012 — Dark mode: etymology role colors

Where: EntitySheet, dark mode

Observation: In dark mode, the green (meaning) and blue (iconic) role colors in etymology are slightly muted against the dark background.

Recommendation: Lighten the role colors by 15–20% in dark mode (e.g. green #4CAF50 → #66BB6A).

Why it helps: Better legibility of the role-color system in dark mode.

Effort: S · **Priority:** Low

UX-013 — Review: add progress indicator

Where: Review session

Observation: The review header shows '1 / 12' but there's no visual progress bar.

Recommendation: Add a thin progress bar under the header that fills as cards are completed.

Why it helps: Gives a sense of progress and motivates completion.

Effort: S · **Priority:** Low

UX-014 — Bigger idea: Stats/progress view

Where: New feature

Observation: There's no way to see learning progress over time — total words learned, review accuracy, streak, etc.

Recommendation: Add a simple stats page accessible from the hamburger menu showing: total saved, per-tier breakdown, review accuracy last 7/30 days, and a simple streak counter.

Why it helps: Motivates continued use. Provides a sense of accomplishment.

Effort: L · **Priority:** Low

UX-015 — Bigger idea: Quick path from word → tree

Where: EntitySheet

Observation: Getting to the decomposition tree requires: open EntitySheet → scroll to etymology → tap ■. Three steps.

Recommendation: Add a small tree icon directly on the saved-shelf card (next to the star). Tapping it opens the tree directly.

Why it helps: Faster access to the most visually impressive feature of the app.

Effort: M · **Priority:** Low

Coverage Matrix

Area	Mobile Light	Mobile Dark	Desktop	Status
Home / Shell	✓	✓	✓	Covered
Search (Dict)	✓	✓ (partial)	✓	Covered
Search (Component)	✓	—	—	Covered
Saved Shelf	✓	✓	✓	Covered
EntitySheet	✓	✓	✓ (modal)	Covered
Status Control	✓	—	—	Covered
Etymology Drill	✓	—	—	Covered
Tree Modal	✓	—	—	Covered
Deep Links	✓	—	—	FAIL
Review Launch	✓	✓	—	Covered
Review Cards	✓	—	—	Partial
FSRS Regression	—	—	—	BLOCKED
Phonetics	✓	✓	—	Covered
Sentence Studio	✓	✓	✓	Covered
Crushed Tabs Regr.	✓	—	—	FIXED ✓
Network Graph	✓	—	✓	Covered
Components Graph	✓	—	✓	Covered
Persistence	✓	—	—	Covered
Back/Forward Nav	✓	—	—	Covered
Console Errors	✓	—	—	0 errors ✓
Auth Modal	✓	—	—	Covered
Palette (/palette/)	—	—	—	Excluded by owner
Cross-device Sync	—	—	—	Out of scope

Appendix: Console Errors & Environment

Console Errors: ZERO uncaught errors across all pages and viewports.

Console Warnings: None observed during testing.

Environment:

- Browser: Chromium headless (Playwright v1.53)
- Mobile viewport: 390×844 (iPhone 14 equivalent), touch enabled, iOS UA
- Desktop viewport: 1280×900, no touch, default UA
- Color schemes: light (prefers-color-scheme: light), dark (prefers-color-scheme: dark)
- Build: chinese v84, built Mon May 11 16:47:15 UTC 2026
- Network/Components versions: network v5, components v2

localStorage Keys Used by App:

- chinese.saved — saved word list with timestamps
- chinese.fsrs.v1 — FSRS spaced repetition schedule data
- chinese.savedSort — current sort preference (recent/pinyin/strokes/hsk/common)
- chinese.reviewSettings — review session settings (enabled drills, shuffle, etc.)